

About the Prediction Monitoring & Revision Standard

What Is PMRS?

The Prediction Monitoring & Revision Standard (PMRS) is the PGA™ Parent Standard governing what happens to a Governed Prediction after it has been formed and entered its operational lifecycle but before its final outcome and predictive performance are assessed.

PMRS establishes how a prediction is monitored as information, assumptions, conditions, feasibility, uncertainty, collective structure, and Operational Reality change over time.

It determines when a prediction may legitimately remain unchanged, when it requires review, when revision becomes necessary, when a newer prediction state supersedes an earlier one, and when a prediction should be withdrawn from current use.

Its governing principle is:

A prediction does not remain governed merely because it was governed once.

Why Does PMRS Exist?

Predictions are created at a particular moment using information and conditions available at that time.

The environment around them continues to change.

New information may emerge.

Assumptions may weaken.

Operational conditions may change.

Feasibility may deteriorate.

Uncertainty may increase.

Collective predictive structures may shift.

The prediction may itself influence the environment it predicts.

Without monitoring and revision governance, a prediction that was legitimate when formed can remain operational long after the conditions supporting it have materially changed.

The central distinction is:

Valid Initial Prediction $\neq$ Valid Continuing Prediction

PMRS exists to govern that difference.

What Does PMRS Govern?

PMRS governs the continuing predictive state of an existing Governed Prediction.

It covers:

prediction monitoring; Prediction Basis change and decay; assumption change; material condition change; probability-relevant change; uncertainty and confidence change; feasibility change; collective prediction change; Predictor–Reality Proximity change; intervening events; prediction-use feedback; reflexive predictive conditions; Review Triggers; Prediction Review; revision thresholds; Prediction Revision; prediction Preservation; supersession; withdrawal; prediction-state transitions; prediction version continuity; historical prediction preservation; stale prediction detection; monitoring and revision failures.

PMRS governs whether the existing prediction should remain the current prediction.

A Prediction Exists in Time

A prediction is not merely a probability attached to an outcome.

It exists within a temporal structure.

At formation:

Prediction State ₀

Later:

Prediction State ₁

Later still:

Prediction State ₂

Information available at each point may differ.

PMRS preserves the relationship between those states.

A prediction that was legitimate on 1 January may require review on 1 March because the environment has materially changed.

That does not retroactively make the January prediction illegitimate.

It means its continuing governance state has changed.

Prediction Monitoring

Prediction Monitoring determines whether conditions relevant to the prediction remain sufficiently comparable to those under which the prediction was formed or last revised.

Monitoring may be:

continuous; periodic; event-triggered; threshold-triggered; condition-triggered; manually initiated; automatically initiated.

The appropriate monitoring intensity depends upon the prediction.

A long-term strategic forecast may require periodic review.

A high-consequence operational prediction may require continuous monitoring.

Prediction Monitoring ≠ Continuous Model Execution

Monitoring asks whether the predictive conditions remain sufficiently stable.

It does not require constant generation of new predictions.

Material Change

Not every new piece of information requires revision.

PMRS distinguishes:

Change

from:

Material Change

A Material Change is one capable of meaningfully affecting the prediction's:

basis; assumptions; probability; uncertainty; confidence; feasibility; predictive conditions; legitimate interpretation; reliance; permitted use.

Change $\neq$ Material Change by Default

This prevents both uncontrolled revision and stale prediction persistence.

Prediction Basis Decay

A Prediction Basis may deteriorate over time.

Data may become stale.

Signals may lose relevance.

Sources may become unreliable.

New information may contradict earlier information.

Operational conditions may move outside the environment represented by the original basis.

PMRS recognizes this as:

Prediction Basis Decay

Prediction Basis Decay does not automatically invalidate the prediction.

It creates a governance question:

Does the existing prediction remain sufficiently supported under current conditions?

If the answer cannot be established, review may be required.

Assumption Change

Predictions frequently depend upon assumptions.

An assumption that was reasonable when the prediction was formed may later become:

Still Supported

Weakened

Changed

No Longer Supported

Unable to Determine

A material assumption change may require Prediction Review.

However:

Assumption Failure $\neq$ Automatic Prediction Falsification

The predicted outcome may still occur.

The governance issue is whether the original prediction remains methodologically supported.

Probability and Uncertainty Change

A prediction may retain the same headline probability while its underlying uncertainty changes materially.

For example:

70% $\rightarrow$ 70%

does not necessarily mean the predictive state remained unchanged.

The second 70% may rest upon:

weaker information; broader uncertainty; changed assumptions; deteriorated feasibility; different conditions.

Therefore:

Same Probability $\neq$ Same Predictive State

PMRS identifies when such changes require renewed governance.

PPUS remains responsible for Probability & Uncertainty governance itself.

Feasibility Change

An outcome that was initially feasible may later become difficult or impossible to achieve.

Likewise, an outcome previously considered infeasible may become feasible because conditions change.

PMRS monitors these transitions.

PFES owns Prediction Feasibility.

PMRS governs whether a change in feasibility materially affects the continuing state of the prediction.

Collective Prediction Change

Collective predictions can change even when their headline number does not.

A market, expert panel, model ensemble, or agent network may continue reporting:

72%

while:

participant composition changes; independence declines;

common information sources increase; influence becomes concentrated; aggregation changes.

Therefore:

Same Collective Number $\neq$ Same Collective Structure

Where those changes are material, PMRS may trigger review.

Prediction and Changing Operational Reality

Predictions exist inside environments that continue evolving.

Operational Reality may change because of events unrelated to the prediction.

It may also change because people or systems act upon the prediction.

The lifecycle can therefore become:

Operational Reality $\rightarrow$ Prediction $\rightarrow$ Communication $\rightarrow$ Reliance $\rightarrow$ Action $\rightarrow$ Changed Operational Reality
 $\rightarrow$ Prediction Review

This creates a feedback loop.

PMRS governs whether that changed environment affects the continuing legitimacy of the prediction.

Reflexive Predictions

Some predictions can influence the reality they predict.

Examples may include:

financial markets; shortages; demand; congestion; elections; public behavior.

A prediction of shortage may cause purchasing behavior that contributes to the shortage.

A failure prediction may trigger preventive action that prevents the failure.

PMRS requires material predictive influence to remain identifiable where it changes the conditions surrounding the prediction.

Prediction Influence $\neq$ Predictive Accuracy

And:

Prediction Influence $\neq$ Outcome Control

Review Triggers

A Review Trigger identifies a condition requiring formal reconsideration of the prediction.

Potential triggers include:

material new information; Prediction Basis Decay; assumption weakening;

assumption failure; feasibility change; material uncertainty change; probability-relevant information; collective-structure change; intervening event; prediction-use feedback; major operational change; Prediction Horizon milestone.

The distinction is critical:

Review Trigger $\neq$ Revision Requirement

A review may conclude that the prediction should remain unchanged.

Prediction Review

Prediction Review asks whether the current prediction remains sufficiently supported.

Possible outcomes include:

Prediction Remains Current

Prediction Remains Current with Qualification

Revision Required

Supersession Required

Withdrawal Required

Additional Governance Required

Unable to Determine

Review is therefore a governance decision point.

It is not an automatic rewriting mechanism.

Review $\neq$ Revision

Prediction Preservation

A prediction should remain unchanged where material evidence does not justify changing it.

This prevents continuous adaptation of predictions merely because new information becomes available.

Preservation is itself a legitimate governance outcome.

A prediction architecture should protect against both:

Under-Revision

and:

Over-Revision

More Frequent Revision $\neq$ Better Prediction Governance

Prediction Revision

Revision occurs when material change requires modification of predictive meaning.

A revision may affect:

probability; uncertainty; range; assumptions; Prediction Basis; feasibility; conditions; Prediction Horizon; predictive scope; qualification.

The revised prediction becomes a new governed state.

The previous prediction remains preserved.

Revision Does Not Rewrite History

This is one of the foundational PMRS boundaries.

Suppose a prediction originally states:

Prediction P1 — 62%

New information later justifies:

Prediction P2 — 81%

The legitimate history is:

P1 → Material Change → Review → P2

It is not:

P1 silently changed into 81%.

The original prediction must remain reconstructable.

Therefore:

Revision ≠ Historical Replacement

Prediction Version Continuity

Predictions may evolve through multiple governed states:

P1 → P2 → P3 → P4

Each state should remain identifiable.

The Prediction Version Chain enables later reconstruction of:

what was predicted; when it was predicted; what information existed; what changed; why revision occurred; which prediction was current; which prediction was communicated or used.

This becomes essential when predictive performance is later evaluated.

Supersession

Supersession occurs when a newer governed prediction replaces an earlier prediction as the current operational prediction.

The earlier prediction remains part of history.

Supersession ≠ Deletion

A superseded prediction may still be necessary for:

audit; historical analysis; decision reconstruction; calibration; predictive performance assessment.

Withdrawal

A prediction may need to be withdrawn where it can no longer legitimately remain active.

Possible reasons include:

invalidated Prediction Basis; major methodological defect; unrecoverable assumption failure; incorrect Prediction Object; serious integrity problem.

Withdrawal ends the prediction's active status.

It does not erase its existence.

Withdrawal $\neq$ Prediction History Deletion

And:

Withdrawal $\neq$ Prediction Failure

The eventual outcome may still occur.

Outcome Knowledge Must Not Rewrite the Prediction

One of the most important functions of PMRS is protecting predictions from hindsight reconstruction.

Once an outcome becomes known, the historical prediction must not be modified to make it appear more accurate.

A prediction cannot legitimately move from:

55% before the outcome

to:

80% in the historical record after the outcome

without preserving the original 55% prediction.

Outcome Knowledge Must Not Rewrite Historical Prediction

This protects the integrity of later predictive-performance assessment.

Revision Gaming

Prediction systems may otherwise be tempted to improve apparent performance through repeated retrospective modification.

Revision Gaming may include:

deleting failed predictions; changing probability shortly before outcome; changing Prediction Horizon; removing failed assumptions; changing the Prediction Object; rewriting earlier prediction states.

PMRS prevents this by requiring historical continuity.

A revised prediction must not erase the prediction that preceded it.

Monitoring Failure

A prediction may have been legitimate when formed but later become stale because material change was not monitored.

That is:

Monitoring Failure

It is not necessarily:

Prediction Error

The distinction allows governance systems to identify where lifecycle governance failed.

Revision Failure

A material change may be correctly detected while the prediction is nevertheless left unchanged.

This may constitute:

Revision Failure

Again:

Revision Failure $\neq$ Initial Prediction Failure

The original prediction may have been legitimate.

The failure occurred later in its lifecycle.

Human, AI, and Autonomous Revision

PMRS can operate across:

human forecasting; institutional forecasting; AI prediction systems; model ensembles; autonomous agents; prediction markets; hybrid Human-AI systems.

AI may identify potential revision triggers.

Autonomous systems may perform revisions where appropriately governed.

But:

AI Revision Recommendation $\neq$ Governed Prediction Revision

and:

Autonomous Revision $\neq$ Historical Rewrite

The same historical continuity requirements remain applicable regardless of who or what performs the revision.

Stale Prediction Propagation

A prediction may be revised at its source while older copies remain active elsewhere.

For example:

Prediction System $\rightarrow$ API $\rightarrow$ Dashboard $\rightarrow$ AI Agent $\rightarrow$ Operational System

The source prediction may become P2 while a downstream agent continues acting upon P1.

PMRS therefore distinguishes:

Available Prediction Copy

from:

Current Prediction State

A distributed prediction ecosystem requires sufficient state continuity to identify stale downstream predictions.

What PMRS Does Not Govern

PMRS does not govern:

original Prediction Object formation; original Predictive Purpose; Prediction Methodology; original Prediction Basis; original Predictive Assumptions; probability calculation; Prediction Feasibility itself; Collective Prediction formation; prediction representation; Prediction Interpretation Boundary; Reliance Legitimacy; Prediction Use Boundary; establishment of Operational Reality; final outcome determination; predictive performance assessment.

PMRS governs the continuing state of the prediction as conditions change.

Cross-Architecture Boundaries

PBAS — Prediction Basis Standard governs what legitimately forms Prediction Basis.

PAS — Prediction Assumption Standard governs Predictive Assumptions.

PPUS — Prediction Probability & Uncertainty Standard governs probability and uncertainty representation.

PFES — Prediction Feasibility Standard governs Prediction Feasibility.

CPGS — Collective Prediction Governance Standard governs legitimate Collective Prediction formation and representation.

PCRS — Prediction Communication & Reliance Standard governs representation, interpretation, reliance, and use.

PMRS — Prediction Monitoring & Revision Standard governs continuing predictive state and legitimate change.

ORA™ — Operational Reality Architecture may establish governed Operational Reality relevant to changing conditions.

ADIT® may govern evidence supporting review or revision.

VALIDOS™ may govern validation relevant to predictive or revision processes.

POFPS — Prediction Outcome & Performance Standard governs subsequent prediction-to-outcome comparison and predictive performance.

Position Within PGA™

PMRS is Parent Standard 10 of 12 within PGA™.

The complete Prediction Governance Flow is:

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

Current Position:

Monitoring & Revision

PMRS sits between:

PCRS — Prediction Communication & Reliance Standard

and:

POFPS — Prediction Outcome & Performance Standard

It governs the period during which the prediction is operationally alive and the world around it continues to change.

Why PMRS Matters

A prediction system without revision governance creates two opposite risks.

The first is:

stale predictions that remain operational after their supporting conditions have changed.

The second is:

predictions that are continuously rewritten until their historical errors disappear.

Both destroy predictive accountability.

PMRS establishes the middle path:

Monitor continuously enough to detect material change. Revise only when governance justifies revision.

Preserve every historically meaningful predictive state.

What Organizations Gain

PMRS enables organizations to establish:

structured prediction monitoring; explicit Review Triggers; controlled prediction revision; version continuity; stale prediction detection; revision traceability; supersession governance; withdrawal governance; distributed prediction-state synchronization; historical prediction integrity; resistance to hindsight manipulation; reliable foundations for later predictive-performance assessment.

The result is not merely a prediction that can change.

It is a prediction whose change itself is governable .

Use Case 1 — Supply Chain Forecast Becomes Stale After a Major External Event

A manufacturer produces a governed prediction:

72% probability of sufficient semiconductor supply for the next six months.

The prediction is legitimately formed using:

supplier capacity; shipping schedules; existing inventory; production commitments; geopolitical conditions.

Three weeks later, a major export restriction affects one of the principal supply regions.

The prediction still displays:

72%

across procurement dashboards.

No outcome has yet occurred.

The critical governance question is:

Does the original 72% prediction remain legitimate merely because its Prediction Horizon has not expired?

PMRS identifies the export restriction as a potential:

Material Intervening Event

The event triggers Prediction Review.

Review determines that:

Prediction Basis materially changed; a key assumption weakened; feasibility conditions changed; the original probability requires reconsideration.

The existing prediction therefore cannot silently continue as current.

A revised prediction is created:

P1 — 72%

→ Material Change

→ Prediction Review

→ P2 — 41%

P1 remains preserved.

P2 becomes the current governed prediction.

Valid Initial Prediction $\neq$ Valid Continuing Prediction

Revision $\neq$ Historical Replacement

Use Case 2 — Preventive Action Changes the Outcome the Prediction Was Measuring

An industrial AI system predicts:

89% probability of critical equipment failure within seven days.

The prediction triggers a governed maintenance intervention.

The component is replaced.

The failure never occurs.

A simplistic performance system later records:

Prediction: Failure — Outcome: No Failure → Prediction Incorrect

But the prediction materially influenced the environment.

The critical governance question is:

Can the prediction be evaluated as though the predicted environment remained unchanged after the prediction was issued?

PMRS preserves the lifecycle:

Prediction → Communication → Reliance → Maintenance Action → Changed Operational Reality

The maintenance intervention becomes a material prediction-use feedback event.

The original prediction remains preserved.

It is not retrospectively rewritten.

The changed operational condition is recorded for subsequent outcome and performance governance under POFPS.

Prediction Influence $\neq$ Predictive Accuracy

Prevented Outcome $\neq$ Automatically Failed Prediction

PMRS therefore preserves the information required for POFPS to distinguish predictive performance from intervention effects.

Canonical Boundary

PMRS governs the continuing lifecycle state of an existing Governed Prediction as information, assumptions, conditions, feasibility, uncertainty, collective structure, prediction use, and Operational Reality change. It determines when monitoring identifies material change, when Prediction Review is required, and when the prediction should remain current, be qualified, revised, superseded, withdrawn, or returned to earlier PGA™ governance while preserving complete historical prediction continuity. PMRS does not determine the eventual outcome or predictive performance.

Canonical Closing Statement

Prediction governance cannot freeze the world at the moment a forecast is made, and it must not rewrite the forecast after the world has changed.

PMRS governs the space between those two failures by preserving both continuing predictive legitimacy and historical predictive truth.

A revised prediction must not erase the prediction that preceded it.

Governed Links

[About Prediction Monitoring & Revision Standard \(PMRS\)](#)

[Prediction Change Monitoring Module \(PCHM\)](#)

[Review Trigger Module \(RTM\)](#)

[Prediction Revision Module \(PRVM\)](#)

[Prediction Supersession Module \(PSSM\)](#)

[Prediction Withdrawal Module \(PWM\)](#)

↑

NON-EXECUTABLE NOTICE

OOF® — OriginOpen® Foundation is a non-executable methodological authority.
This website and all publications are reference only.

CANONICAL LANGUAGE NOTICE

UCL™ is the canonical language layer of OOF® publications.
English is its carrier language.
In case of ambiguity, the published English HTML text prevails.

RIGHTS & PROTECTION

© OOF® — OriginOpen® Foundation
Protected under MIP® — Methodological Intellectual Property.
All rights reserved.



contact@originopenfoundation.org